Week 5 Practical: RBAC, NSG & Storage Security

1. Role-Based Access Control (RBAC)

Explanation: RBAC roles were assigned at Resource Group scope to enforce least privilege.

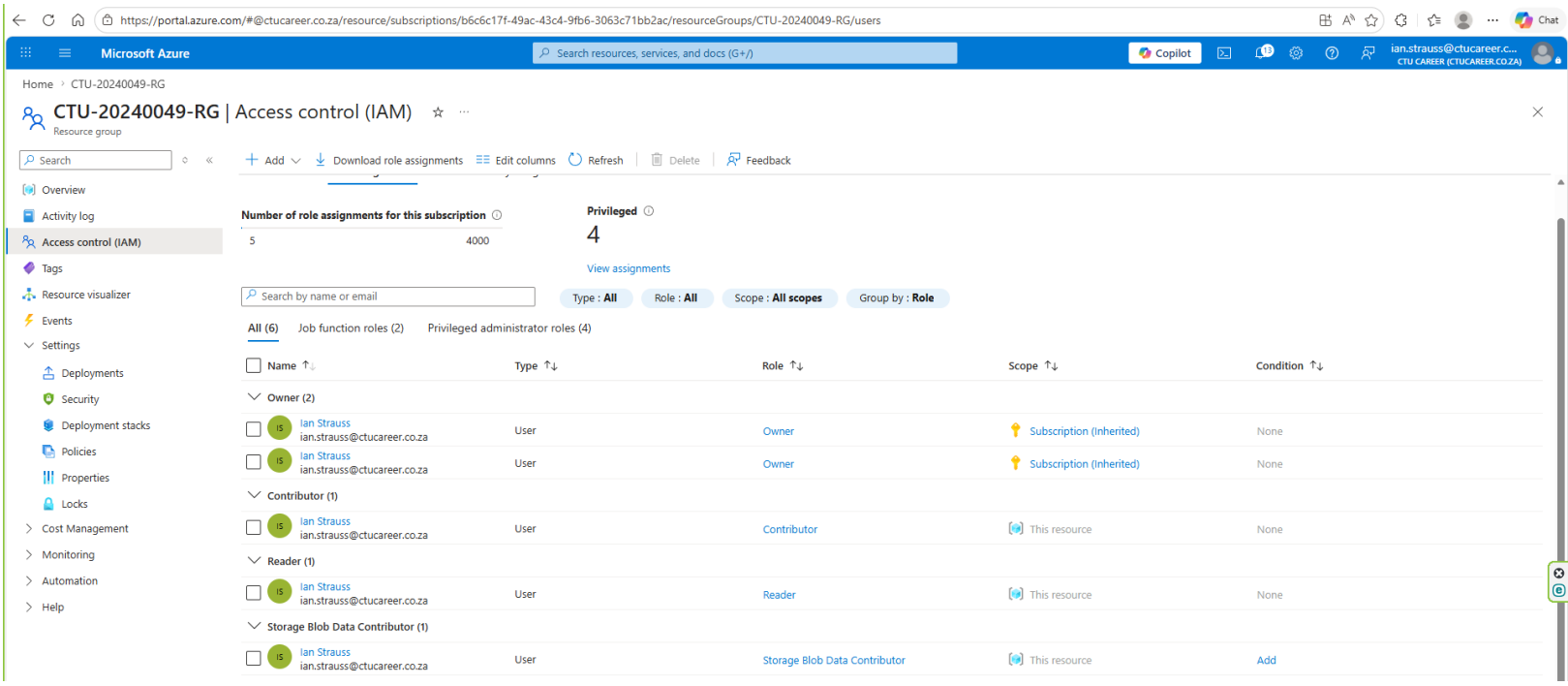
Roles assigned:

- Contributor → user-ops (or substitute account)

- Reader → user-audit (or substitute account)

- Storage Blob Data Contributor → user-data (or substitute account)

Screenshot:



Scope explanation: Resource Group scope ensures users only access assigned resources, not the entire subscription.

2. Network Security Group (NSG)

NSG Name: NSG-yourstudentnumber

Inbound Rules:

- 3389 (RDP) or 22 (SSH) → Allow (Remote access)

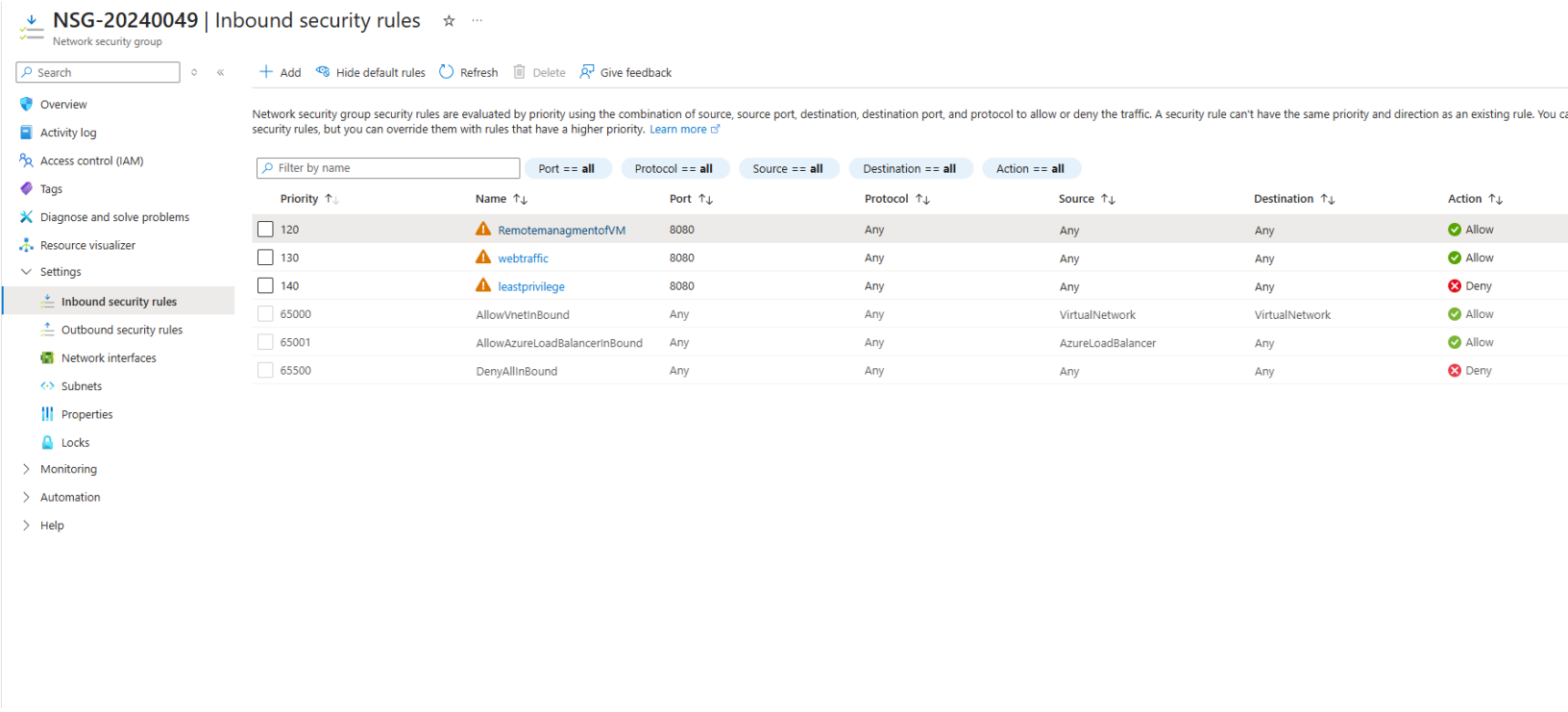
- 80 / 443 → Allow (Web traffic)

- Any → Deny (Block all other inbound traffic)

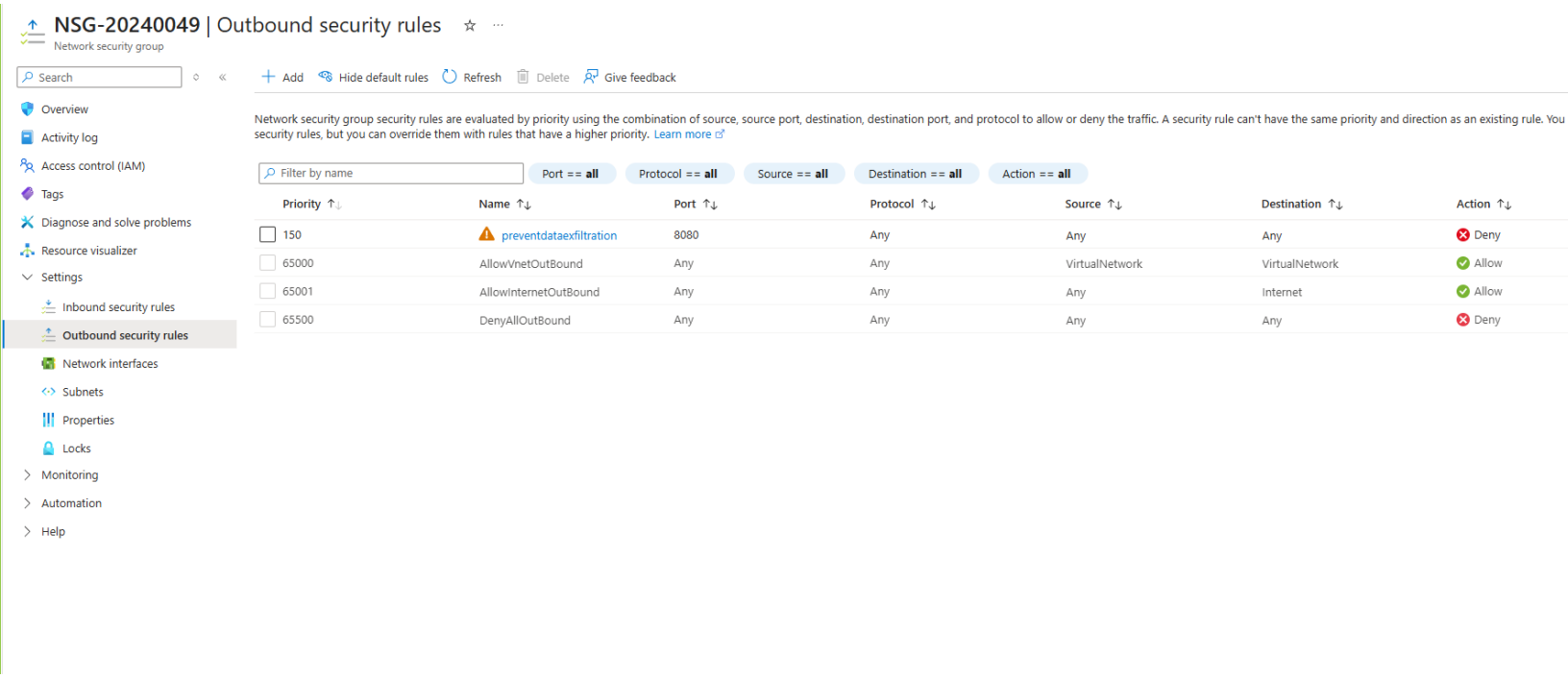
Outbound Rules:

- Internet → Deny (Prevent data exfiltration)

Screenshot:



Screenshot:



Screenshot: [INSERT NSG SUBNET ASSOCIATION HERE]

Explanation: NSG rules control traffic flow and enforce network-level security policies.

3. Storage Account & Containers

Storage Type: StorageV2 (General Purpose v2)

Replication: LRS (Locally Redundant Storage)

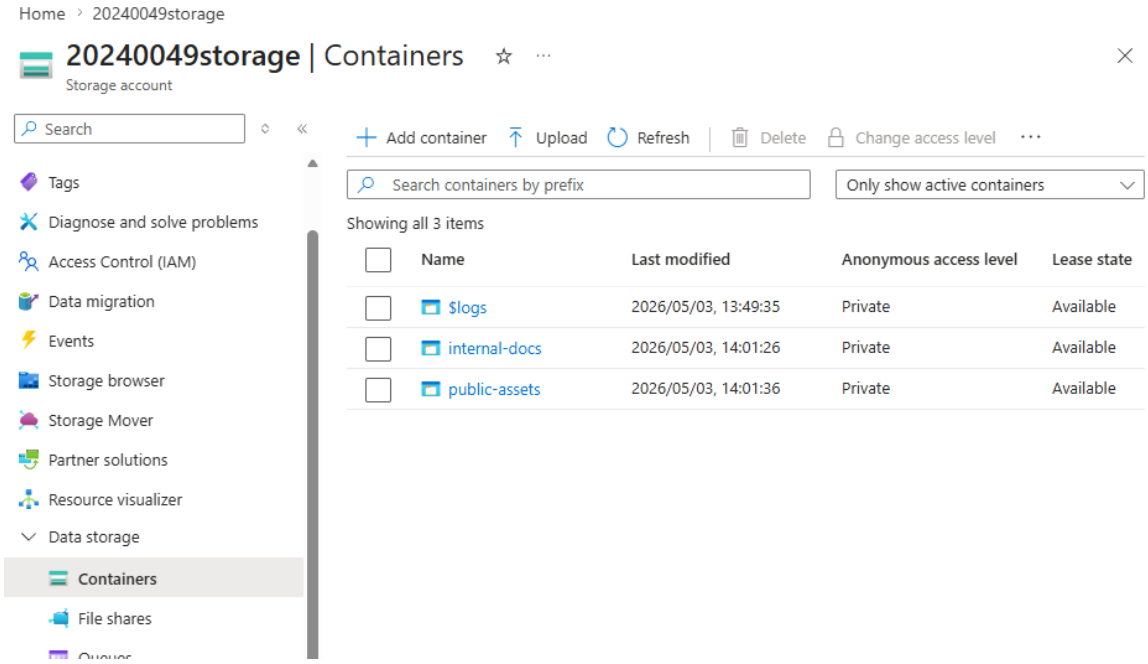
Access Tier: Hot

Containers:

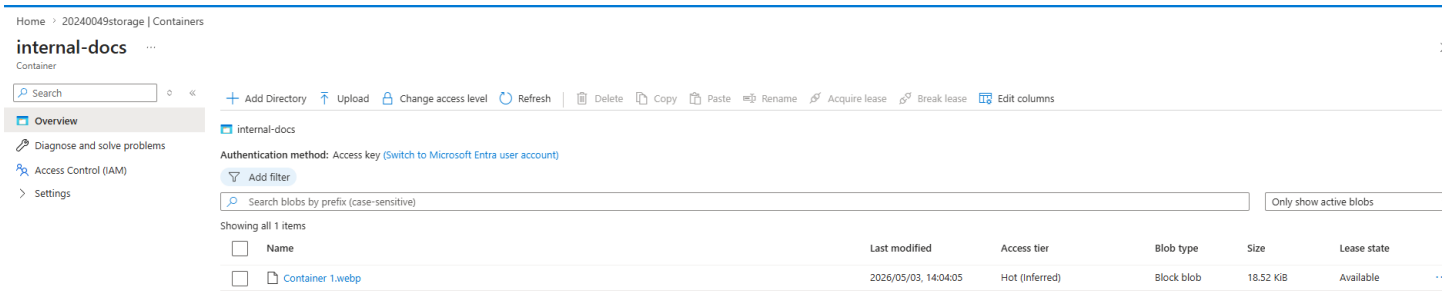
- internal-docs (Private)

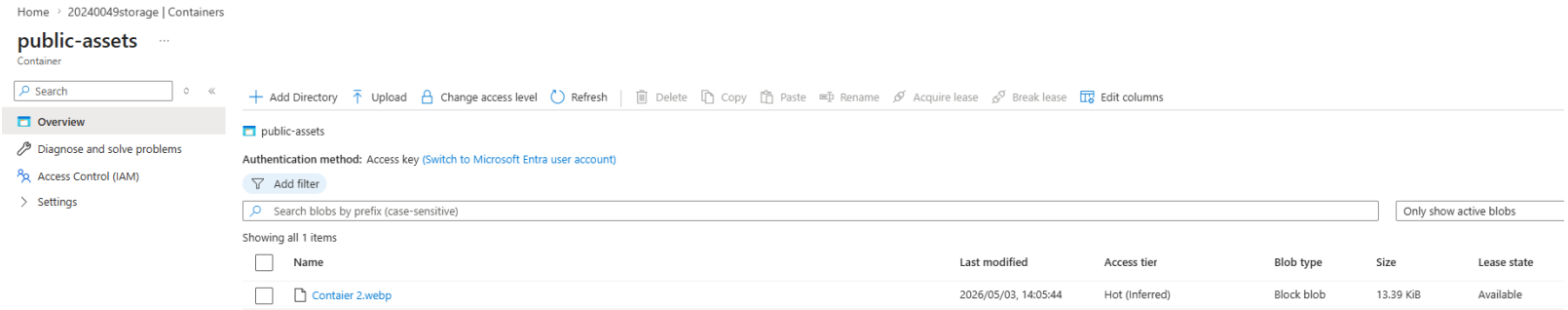
- public-assets (Private)

Screenshot:



Screenshot:



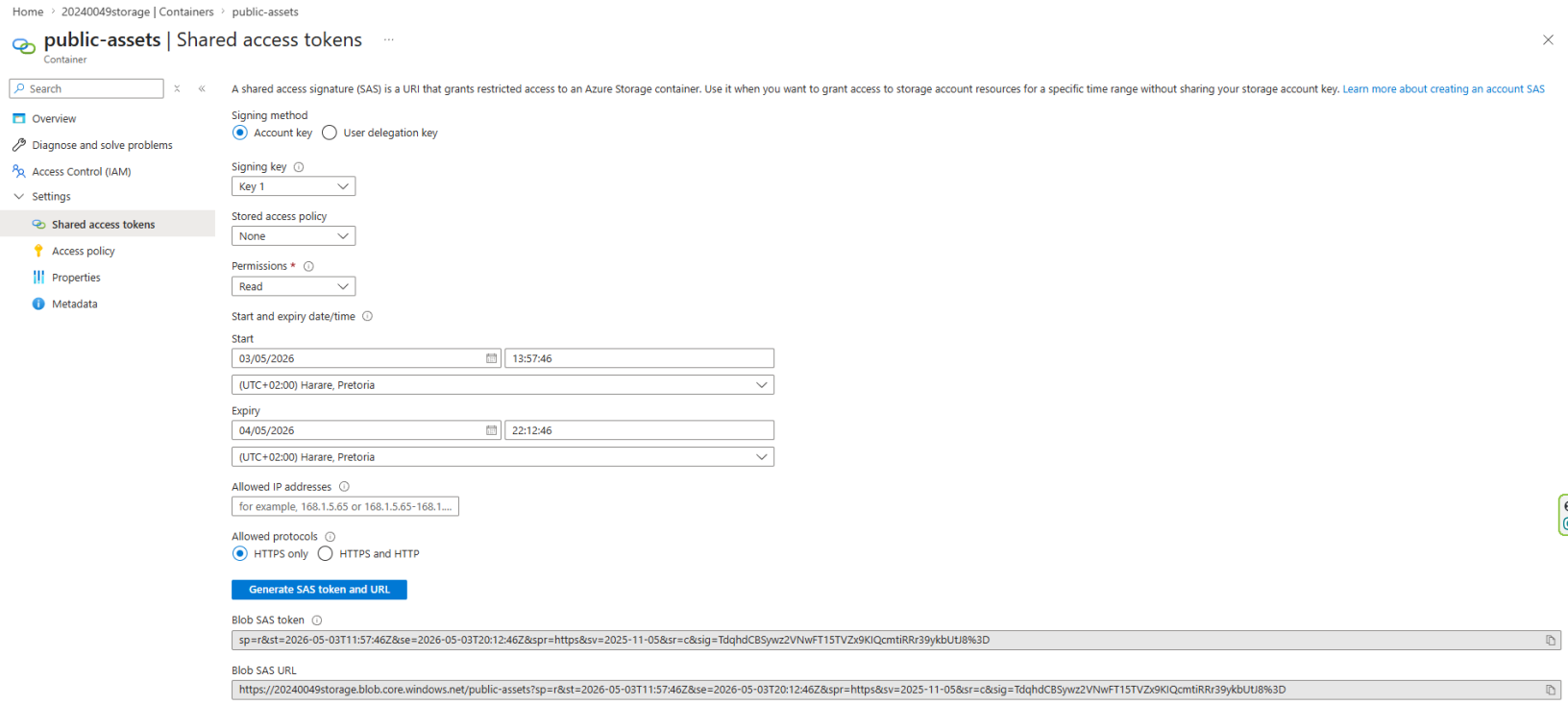


4. SAS Token & Access Control

A SAS token was generated for temporary read-only access.

Expiry: 24 hours

Permissions: Read-only



Security Note: SAS tokens provide direct access without authentication and must be protected carefully.

5. Conclusion

This practical implemented RBAC, NSG, and Storage security to enforce layered cloud security principles.